

# Kanishk Jain

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## EDUCATION

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### Université de Montréal | Mila

Montréal, Canada

PhD in Computer Science and Engineering; **GPA: 4.3**

Sep 2023 – Present

- Research Focus: Reinforcement Learning, Multi-Agent Frameworks, Foundational Multimodal Models, Lifelong Learning, Multimodal Representation Learning
- Advisors: Prof. Sarath Chandar, Prof. Ross Goroshin

### International Institute of Information Technology

Hyderabad, India

MS by Research in Computer Science and Engineering; **GPA: 9.33/10**

Aug 2021 – Dec 2022

- Research Focus: Multimodal Learning, Visual Grounding, Language-Guided Autonomous Navigation, Semi-Supervised Learning
- Advisor: Dr. Vineet Gandhi

## RESEARCH PUBLICATIONS

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### Discovering Failure Modes in Vision-Language Models using RL

Paper

**Kanishk Jain**, Qian Yang, Shravan Nayak, Parisa Kordjamshidi, Nishanth Anand, Aishwarya Agrawal *arXiv 2026*

- An RL framework that trains a questioner agent to generate increasingly complex queries, automatically exposing weaknesses in vision-language models without human intervention.

### Prioritized Concept Learning via Relative Error-driven Sample Selection

Paper

Shivam Chandhok, Qian Yang, Oscar Manas, **Kanishk Jain**, Leonid Sigal, Aishwarya Agrawal *CVPR 2026*

- A data-efficient training framework for VLMs that dynamically selects samples based on model's learning propensity

### Benchmarking Vision Language Models for Cultural Understanding

Paper

Shravan Nayak, **Kanishk Jain**, Rabiul Awal, Siva Reddy, et al.

*EMNLP 2024 (Oral)*

- Introduced CulturalVQA, a benchmark for evaluating Vision Language Model's understanding of geo-diverse cultural elements, revealing performance disparities across regions and cultural facets.

### Test-Time Amendment with a Coarse Classifier for Fine-Grained Classification

Paper

**Kanishk Jain**, Shyamgopal Karthik, Vineet Gandhi

*NeurIPS 2023*

- Proposed Hierarchical Ensembles (HiE), a post-hoc correction strategy that utilizes coarse-grained predictions at test-time to significantly reduce mistake severity and enhance fine-grained classification accuracy

### Instance-Level Semantic Maps for Vision Language Navigation

Paper

Laksh Nanwani, Anmol Agarwal, **Kanishk Jain**, et al.

*ROMAN 2023*

- Developed Instance-Level Semantic Maps to resolve object ambiguity, enabling navigation agents to precisely ground complex natural language instructions.

### Ground then Navigate: Language-guided Navigation in Dynamic Scenes

Paper

**Kanishk Jain\***, Varun Chhangani\*, Amogh Tiwari, K. Madhava Krishna, Vineet Gandhi

*ICRA 2023*

- Developed a visual-grounding-based planning framework that maps language instructions to navigable road regions, enabling navigation in dynamic outdoor environments.

### Bringing Generalization to Deep Multi-View Pedestrian Detection

Paper

Jeet Vora, Swetanjal Dutta, **Kanishk Jain**, Shyamgopal Karthik, Vineet Gandhi

*WACV-W 2023*

- Developed an evaluation framework and synthetic dataset to benchmark generalization in multi-view pedestrian detection to unseen configurations and realistic scenarios.

### Comprehensive Multi-Modal Interactions for Referring Image Segmentation

Paper

**Kanishk Jain**, Vineet Gandhi

*ACL Findings 2022*

- Developed a Transformer-based architecture for Referring Image Segmentation, leveraging synchronous multi-modal attention and hierarchical aggregation to enhance segmentation performance.

### Grounding Linguistic Commands to Navigable Regions

Paper

**Kanishk Jain\***, Nivedita Rufus\*, Unni Krishnan\*, Vineet Gandhi, K. Madhava Krishna

*IROS 2021*

- Introduced Referring Navigable Regions (RNR), a novel task for grounding linguistic commands to navigable road regions, enabling autonomous agents to follow human instructions for navigational maneuvers.

## WORK EXPERIENCE

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### Noah's Ark Lab | Huawei

*Intern Researcher*

*Sep 2025 – Jan 2026*

- Developing modality-invariant foundation models to mitigate modality competition by leveraging contrastive learning for robust multimodal representations.

### Université de Montréal | IFT 6765

*Teaching Assistant*

*Jan 2025 – Apr 2025*

- Held weekly office hours, designed and graded assignments, and mentored teams on project ideas.

### Université de Montréal | Mila

*PhD Student*

*Sep 2023 – Present*

- Developing a Reinforcement Learning framework that automatically discovers failure modes in Vision-Language Models, training an adversary to generate inputs that expose systematic blind spots and produce targeted data for robustness in safety-critical applications.
- Exploring early-exit methods for streaming video understanding to enable real-time decision making with reliability guarantees.

### CVIT, IIIT Hyderabad

*Research Engineer*

*Mar 2023 – Aug 2023*

- Developed Hierarchical Ensembles to reduce mistake severity in fine-grained classification, improving performance in supervised and semi-supervised settings.
- Designed and deployed a real-time player tracking solution in Bird's Eye View, currently utilized in live cricket broadcasts.

### CVIT, IIIT Hyderabad

*Research Assistant*

*Aug 2021 – Dec 2022*

- Led research on language-guided autonomous navigation, leveraging visual grounding for planning and human-interpretable decision-making.
- Enhanced multi-modal interaction models for referring image segmentation, increasing the efficiency and precision of visual-linguistic model performance.
- Developed an analytics tool for Counter-Strike: Global Offensive (CSGO) games, providing actionable insights and improving gameplay strategies for competitive teams.

## SELECTED PROJECTS

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**T2I Generation using SSMs:** Analyzing the effectiveness of Selective State Space Models (SSM) in reducing computational complexity for text-to-image generation.

**Top-View Player Tracking:** Developed and deployed a real-time player tracking solution in Bird's Eye View, used live during the 2022 Asia Cup Cricket Tournament.

**Stereo SLAM:** Created 3D point clouds from stereo images and estimated motion/pose by applying the iterative Perspective-n-Point (PnP) algorithm for 2D-3D correspondences.

**Pose Graph Optimization:** Implemented the Levenberg–Marquardt algorithm to optimize robot poses using Odometry and Loop Closure constraints for 1D and 2D Simultaneous Localization and Mapping (SLAM).

**Unity Game for Amblyopia:** Developed a Unity-based game for diagnosing Amblyopia, controlled through eye gaze movements tracked by an eye-tracking device.

**Neuro Rehab Systems:** Designed a rehabilitation tool aimed at assisting recovery from nervous system injuries and minimizing functional impairments.

## TECHNICAL STRENGTHS

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**Languages:** Python, Java, C++, C#, Node JS, Javascript

**Frameworks:** PyTorch, Keras, Tensorflow, OpenCV

**Tools:** CARLA, Unity 3D, TensorRT, Open3D, Matlab

## RELEVANT COURSES

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**ML/AI Courses:** Statistical Methods in AI, Computer Vision, Mobile Robotics, Topics in Optimization Methods, Topics in ML, Cognitive Science and AI

**Core Science:** Computer Programming, Operating Systems and Algorithms, Data Structures

**Other Courses:** Digital Image Processing, Digital Signal Processing, Linear Algebra, Probability and Random Processes, Discrete Mathematics

## ACADEMIC SERVICE

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**Reviewer:** NeurIPS (2025, 2026); CVPR (2024); WACV (2024, 2025); ACL (2024); ECCV (2024, 2026); ICRA (2024).

## ACHIEVEMENTS

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**Morocco Solidarity Hackathon:** Member of the winning team; developed a solution to predict human trafficking behavior from social media posts.

**Qualcomm Innovation Fellowship:** Led the winning team of the Qualcomm Innovation Fellowship (QIF) 2020 in India.

**JEE Mains:** Ranked in the top 0.2% nationally among 1,200,000 candidates in JEE Mains.

**JEE Advanced:** Secured a rank of 4539 among 150,000 candidates in JEE Advanced.

## REFERENCES

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**Dr. Vineet Gandhi, Dr. K. Madhava Krishna**